

| Question |     |       |  | Marking details                                                                                                        | Marks Available |          |          |           |          |      |
|----------|-----|-------|--|------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|------|
|          |     |       |  |                                                                                                                        | AO1             | AO2      | AO3      | Total     | Maths    | Prac |
| 3<br>FT  | (a) |       |  | -101 (1)<br>1.14 (1)<br>53 (1)                                                                                         |                 |          | 3        |           | 0        |      |
|          | (b) | (i)   |  | Boiling points of halogens increase (down the group) (1)<br>No not the same (1)                                        |                 |          | 2        |           |          |      |
|          |     | (ii)  |  | fluorine (1)<br>chlorine (1)                                                                                           |                 | 2        |          |           |          |      |
|          |     | (iii) |  | bromine                                                                                                                |                 | 1        |          |           |          |      |
|          |     | (iv)  |  | increases                                                                                                              | 1               |          |          |           |          |      |
|          |     | (v)   |  | Ticks in second, third and fourth boxes.<br><br>4 boxes ticked max (2)<br>5 boxes ticked max (1)<br>6 boxes ticked (0) |                 | 3        |          |           |          |      |
|          | (c) |       |  | Deep shaft mining                                                                                                      | 1               |          |          |           |          |      |
|          |     |       |  | <b>Question 3 total</b>                                                                                                | <b>2</b>        | <b>6</b> | <b>5</b> | <b>13</b> | <b>0</b> |      |